

DESN2000 T11D A — Smart Cottage Hub

This project is a local, wired smart-home prototype for a holiday cottage. It runs on an **LPC2478 / ARM7TDMI-S** with a QVGA touchscreen and Daughter Board.

The system uses time, ambient light and user input to simulate:

- two automated blinds;
- an espresso-machine smart plug;
- house-light status;
- heating/cooling status;
- a doorbell song with an LED Disco effect; and
- Do Not Disturb with a missed-guest counter.

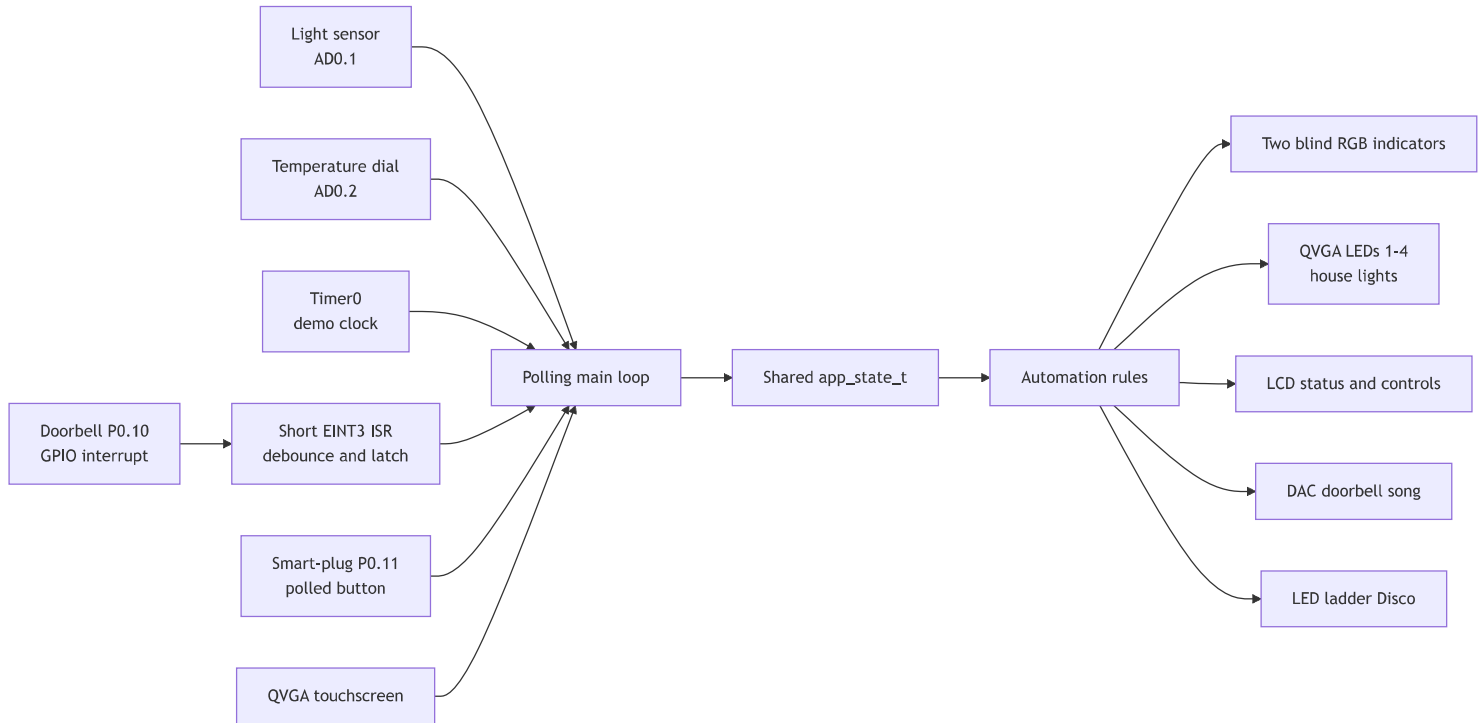
No real 240 V appliance is controlled. The project uses LEDs, the LCD and other low-voltage board outputs as safe simulations.

Current status

- The integrated Keil project builds and links with **0 errors and 0 warnings**.
- Output image: `output/desn2000_t11d_a.axf` .
- Port 0/1 GPIO access is consistently configured for Fast GPIO, matching the Lab 4 projects and NyanSim's daughter-board button interface.
- The `HOUSE LIGHT` touchscreen control drives QVGA Base Board LEDs 1-4 through the PCA9532 and retains its manual state until `AUTO RESET` is pressed.
- The P0.10 doorbell input is interrupt-driven through the LPC2478 EINT3/VIC source; its ISR debounces and latches the event for the main loop.
- The standalone AD0.1 light-sensor path has been tested successfully in NyanSim.
- Full physical-board testing is still required for the touchscreen, TEMT6000 calibration, interrupt doorbell, house-light LEDs, RGB LEDs and LCD recovery after Disco mode.

Build success proves that the source compiles and links. It does not prove that every hardware feature has passed a board test.

System overview



The application uses a single-threaded **polling superloop** for normal work. There is no RTOS or dynamic memory. The one application interrupt is the P0.10 doorbell GPIO ISR; it performs no audio, LCD or application-state work and only latches a debounced event for the superloop. All current values and output states are stored in one `app_state_t` structure.

Main control loop

Each loop performs the following steps:

1. Read Timer0 and calculate the accelerated demonstration time.
2. Read the light sensor and temperature dial through ADC0.
3. Apply the time, light and temperature rules.
4. Poll the smart-plug button and consume any doorbell event latched by the P0.10 GPIO interrupt.
5. Read one touchscreen command.
6. Update the blind RGB outputs.
7. Refresh the LCD every 250 ms.

Manual controls have priority over automatic rules. `AUTO RESET` clears the manual overrides and recalculates the outputs using the current time and sensor readings.

Automation rules

The demonstration starts at **05:30**. One real second advances the clock by one demonstration hour. The first update therefore changes the display from 05:30 to 06:30.

Demonstration time	Routine	Smart plug	Automatic blinds	House lights
00:00–05:59	Night	Off	Both down	On
06:00–11:59	Morning	On	Both up	Off
12:00–14:59	Day / midday	Off	Blind 1 light-controlled; Blind 2 up	Off
15:00–17:59	Day / afternoon	Off	Blind 1 up; Blind 2 light-controlled	Off
18:00–21:59	Evening	Off	Both down	On
22:00–23:59	Night	Off	Both down	On

Light control

The light sensor produces a 10-bit ADC value from 0 to 1023.

Figure 1 in the Project Brief has north pointing toward the bottom of the page. Blind 1 is therefore north-facing and Blind 2 is west-facing. The north-facing Blind 1 responds to light around midday; the west-facing Blind 2 responds in the afternoon. The other blind stays up to admit useful daylight.

For whichever orientation is active during that time window:

Raw ADC value	Light level	Active blind state	RGB colour
0–300	Dark	Up	Red
301–699	Medium	Mid-way	Green
700–1023	Bright	Down	Blue

The 300 and 700 thresholds are demonstration values. They are not calibrated lux measurements.

During Evening and Night, the time rule closes both blinds regardless of the light reading. A manual touch command continues to override this automatic orientation strategy until `AUTO RESET` is pressed.

Heating and cooling display

The onboard potentiometer represents a simulated room temperature from **16 °C to 30 °C**. The target is 22 °C with a 1 °C deadband.

Simulated temperature	LCD state
16–20 °C	Heating
21–23 °C	Comfortable
24–30 °C	Cooling

This is an LCD demonstration only. There is no real temperature sensor, heater or cooler actuator.

User controls

Input	Action
P0.10 / S1	Doorbell
P0.11 / S2	Toggle smart plug and enter manual mode
Touch <code>PLUG</code>	Toggle smart plug and enter manual mode
Touch <code>BLIND</code>	Cycle both blinds through Up, Mid-way and Down
Touch <code>HOUSE LIGHT</code>	Toggle QVGA LEDs 1-4 and enter manual mode
Touch <code>DND</code>	Toggle Do Not Disturb
Touch <code>AUTO RESET</code>	Clear overrides, DND and event counters, then return to automatic control

When DND is off, a doorbell press plays the song and LED pattern. When DND is on, the song and LEDs are suppressed and the missed-guest counter increases.

The touchscreen only generates one command per press. The user must release the screen before the same control can be triggered again.

House light

The former `LIGHT TEST` control is now `HOUSE LIGHT`. It is independent of the AD0.1 light-sensor input and does not step or simulate the sensor reading.

- Tapping `HOUSE LIGHT` toggles `house_lights_on` and enters manual mode.
- `house_lights_show()` sends the output through I2C0 to the PCA9532.
- QVGA Base Board red LEDs 1-4 turn on or off together; LEDs 5-8 are untouched.
- The manual state is retained across normal time and sensor updates.
- `AUTO RESET` clears the manual override and reapplies the current routine.
- Automatic mode turns the house lights on during Evening/Night and off during Morning/Day.

Interrupt doorbell

P0.10/S1 is an active-high doorbell input configured for a rising-edge GPIO interrupt. LPC2478 Port 0 GPIO interrupts share the EINT3/VIC source 17.

The ISR reads the free-running Timer0 counter for a 100 ms debounce interval, latches a press event, clears the GPIO interrupt and acknowledges the VIC. The main loop consumes that event and owns all longer work: updating the doorbell state, applying DND, playing the song/Disco sequence and recovering the LCD. This keeps audio and display operations out of interrupt context.

Important hardware mapping

Pin/resource	Use
P0.10	Doorbell button; rising-edge GPIO interrupt through EINT3/VIC source 17
P0.11	Smart-plug override button
P0.24 / AD0.1	TEMT6000 light-sensor input
P0.25 / AD0.2	Potentiometer used as simulated temperature
P0.26 / AOUT	DAC speaker output
P0.27 / SDA0, P0.28 / SCL0	PCA9532 control for QVGA LEDs 1-4 house-light output

Pin/resource	Use
P0.22	Active-high LED-ladder enable
P2.1–P2.8	LCD interface and temporary Disco LED-ladder output
P3.16–P3.18	Blind 1 RGB: red=up, green=mid-way, blue=down
P3.19–P3.21	Blind 2 RGB: red=up, green=mid-way, blue=down
SPI0 / P0.20 CS	TSC2046 touchscreen
External SDRAM	LCD framebuffer at 0xA0000000

For a physical TEMT6000 test, connect the sensor circuit to 3.3 V, P0.24/AD0.1 and ground through the supplied 10 kΩ arrangement. Disconnect the J12 accelerometer-Y route from P0.24 first.

The temperature dial uses P0.25/AD0.2. Confirm that J11 selects the potentiometer and that J12 does not also connect the accelerometer-X signal.

Key technical details

ADC

- ADC0 is read by polling; ADC interrupts are disabled.
- AD0.1 reads the light sensor.
- AD0.2 reads the temperature dial.
- Each conversion has a polling timeout and returns the previous valid value if the conversion does not finish.

GPIO

- Startup sets `SCS.GPIOM=1` before entering C, selecting Fast GPIO for Ports 0 and 1.
- Buttons, LCD control, touchscreen chip select, LED-ladder enable and the P1.5 SD/MMC power-cycle output all use the matching 32-bit `FIOXDIR` , `FIOXPIN` , `FIOXSET` and `FIOXCLR` registers.
- P0.10 is an active-high, rising-edge GPIO interrupt routed through the EINT3 VIC source. Its ISR performs the 100 ms Timer0-based lockout and only latches an event; the main loop owns the chime, state update and LCD work.

- P0.11 remains an active-high polled input with software rising-edge detection and a 100 ms lockout.

Timer and audio

- CPU clock: 72 MHz.
- Peripheral clock: 36 MHz.
- Timer0 is configured as a free-running 1 μ s counter.
- The demo clock, both button lockouts and DAC timing share Timer0.
- The doorbell detects its rising edge in a GPIO ISR; the smart-plug button detects its rising edge by polling. Both use a 100 ms lockout.
- Audio is produced as a square wave through the P0.26 DAC.

LCD and touchscreen

- Display size: 240 × 320 pixels.
- Colour format: 16-bit RGB565.
- The framebuffer is stored in external SDRAM.
- The project uses the 16-bit SDRAM path so that P3.16–P3.21 remain available for the blind RGB outputs.
- The TSC2046 touchscreen is read by polling SPI0.
- Touch orientation and pressure thresholds still require physical-board calibration.

Doorbell Disco

The doorbell plays the first 58 notes of the Lab 5 Nyan Cat data. Its nominal audio duration is approximately **3.33 seconds**.

P2.1–P2.8 are shared by the LCD interface and the Daughter Board LED ladder. Disco mode therefore:

1. saves the LCD pin and controller configuration;
2. stops the LCD controller;
3. changes P2.1–P2.8 to GPIO;
4. plays the song and 16-step LED pattern;
5. disables and clears the LED ladder;
6. restores the LCD configuration;
7. reinitialises the LCD panel and redraws the screen; and
8. returns SPI0 to touchscreen mode.

The full event, including LCD recovery, blocks the main polling loop for about four seconds.

Project structure

Path	Purpose
src/main.c	Initialisation and main polling loop
src/app_state.h	Shared application state and enums
src/app_logic.c	Automation and override rules
src/config.h	Thresholds, timing and touch settings
src/inputs/	ADC, buttons and touchscreen drivers
src/helper/	Timer0, DAC audio and blind RGB control
src/active_house_operations/	Display, routines, DND, HVAC and Disco
src/lcd/	Reused Lab 6 LCD, graphics and SDRAM support